

- **INNER JOIN** – Retrieve employees along with their department name and the projects they work on by joining EMPLOYEE, DEPARTMENT, and PROJECT tables.

SELECT

e.Fname AS Employee_First,
e.Lname AS Employee_Last,
d.DName AS Department_Name,
p.PName AS Project_Name,
p.City AS Project_City

FROM EMPLOYEE e

INNER JOIN DEPARTMENT d ON e.DNum = d.DNUM

INNER JOIN PROJECT p ON d.DNUM = p.DNum

ORDER BY d.DName, e.Lname;

Script Output x

Query Result x

SQL | All Rows Fetched: 5 in 0.007 seconds

	EMPLOYEE_FIRST	EMPLOYEE_LAST	DEPARTMENT_NAME	PROJECT_NAME	PROJECT_CITY
1	Sara	bndf	Education	Edu in Oman	Muscat
2	Nabhan	Allamki	Engineering	Electricity mesuring	Barka
3	Mariya	Allamki	IT	DML Implelemntation	Muscat
4	Mohammed	LL	Pharmacy	Brewing coffee	rustaq
5	Razan	Alharthi	Science	Telecommunication	Salalah

- **LEFT JOIN** – Display all employees with their department information and dependent details, including employees who have no dependents, by joining EMPLOYEE, DEPARTMENT, and DEPENDENT tables.

SELECT

e.Fname AS Employee_First,
e.Lname AS Employee_Last,
d.DName AS Department_Name,
dp.Dependent_Name,
dp.Gender AS Dependent_Gender

FROM EMPLOYEE e

INNER JOIN DEPARTMENT d ON e.DNum = d.DNUM

LEFT JOIN DEPENDENT dp ON e.SSN = dp.SSN

ORDER BY e.Lname;

Script Output x

Query Result x

SQL | All Rows Fetched: 5 in 0.007 seconds

	EMPLOYEE_FIRST	EMPLOYEE_LAST	DEPARTMENT_NAME	DEPENDENT_NAME	DEPENDENT_GENDER
1	Razan	Alharthi	Science	Buthaina Alkindi	F
2	Nabhan	Allamki	Engineering	Omer Allamki	M
3	Mariya	Allamki	IT	(null)	(null)
4	Mohammed	LL	Pharmacy	Mariya Allamki	F
5	Sara	bndf	Education	Fatma Almamari	F

- **RIGHT JOIN** – Show all projects with their assigned department and the employees working on them, including projects with no assigned employees, by joining PROJECT, DEPARTMENT, and EMPLOYEE tables.

SELECT

p.PNumber AS Project_No,
p.PName AS Project_Name,
d.DName AS Controlling_Dept,
e.Fname AS Employee_First,
e.Lname AS Employee_Last

FROM EMPLOYEE e

RIGHT JOIN DEPARTMENT d ON e.DNum = d.DNUM

RIGHT JOIN PROJECT p ON d.DNUM = p.DNum

ORDER BY p.PNumber;

Script Output x

Query Result x

- **SELF JOIN** – Retrieve each employee along with their supervisor's name and department by joining the EMPLOYEE table with itself alongside the DEPARTMENT table.

SELECT

```
emp.Fname AS Worker_First,
emp.Lname AS Worker_Last,
mgr.Fname AS Supervisor_First,
mgr.Lname AS Supervisor_Last,
d.DName AS Working_Department
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FROM EMPLOYEE emp

LEFT JOIN EMPLOYEE mgr ON emp.Supervisor = mgr.SSN

INNER JOIN DEPARTMENT d ON emp.DNum = d.DNUM

ORDER BY d.DName, emp.Lname;

Script Output x Query Result x					
SQL All Rows Fetched: 5 in 0.005 seconds					
	WORKER_FIRST	WORKER_LAST	SUPERVISOR_FIRST	SUPERVISOR_LAST	WORKING_DEPARTMENT
1	Sara	bndf	(null)	(null)	Education
2	Nabhan	Allamki	(null)	(null)	Engineering
3	Mariya	Allamki	Razan	Alharthi	IT
4	Mohammed	LL	(null)	(null)	Pharmacy
5	Razan	Alharthi	(null)	(null)	Science